

Maryana Shnitser

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Research Interests: Applied geospatial analysis, spatial economics, climate impacts and adaptation, land-use dynamics and planning

EDUCATION

The George Washington University

Washington, DC

B.A. in Geography, Economics (departmental honors), *summa cum laude* | GPA: 3.89

May, 2025

Minor: Geographic Information Systems (GIS)

Relevant Coursework: Techniques of Spatial Analysis, Geospatial Programming I & II, Open Source GIS, Geovisualization & Cartography, Remote Sensing, Digital Image Processing & Analysis, Econometrics, Intermediate Micro & Macroeconomic Theory, Environmental Economics, Financial Economics

TECHNICAL SKILLS

Data Analysis: Python, R, Stata, SQL, Microsoft Suite (Excel, Word, PowerPoint)

Geospatial Analysis & Cartography: ESRI software (ArcGIS Pro, ArcGIS Online, ArcPy), Adobe Illustrator, Google Earth Engine (GEE), ERDAS Imagine, QGIS, OSM, JOSM, GPS

Languages: English (Native), Ukrainian (Native/Bilingual), Russian (Basic)

RESEARCH EXPERIENCE

George Washington University

Washington, DC

Research Assistant - Rapid Arctic Environmental Change (RACE) Project

February 2024 - May 2026

- Developed GIS workflows to compile, clean, and integrate infrastructure datasets with climate model projections under various climate scenarios, enabling spatial assessment of climate-related risks
- Analyzed economic datasets and modeled ground subsidence and coastal erosion risks to estimate the potential infrastructure damage and land-loss costs to Arctic communities and industries, supporting adaptation planning and future development decisions
- Conducted literature reviews to inform new and ongoing projects, assisted with book chapter and research paper writing, and contributed to grant proposal development for research funding

Field Research Assistant - Circumpolar Active Layer Monitoring (CALM) Project

August 2023 - August 2025

- Collected field data on permafrost active layer thickness at multiple field sites in Alaska, contributing to long-term IPCC datasets and supporting ongoing research on Arctic environmental change
- Supported data management, quality control, and field logistics for Arctic research teams, integrating field observations into standardized datasets used for long-term climate modeling

George Washington University — Department of Economics

August 2024 - December 2024

Economics Thesis, Title: The Impact of Metro Accessibility on Residential Property Values and Population

Clustering: A Spatial Analysis of Washington, DC

- Constructed a comprehensive spatial dataset combining property values, demographic attributes, neighborhood characteristics, and location attributes for residential properties in Washington, DC
- Performed an Ordinary Least Squares (OLS) regression and Geographically Weighted Regression (GWR) to analyze and compare results when accounting for spatial heterogeneity in housing markets
- Applied the Getis-Ord G_i^* statistic to identify population clustering, reveal preference, and analyze market dynamics to inform future policy and planning decisions

PUBLICATIONS, PRESENTATIONS & AWARDS

Streletskiy, D. A., & Shnitser, M. (2025). Cost of permafrost degradation and land use impacts of infrastructure development. In *The Economy of the North – ECONOR 2025* (Chapter 10, p. 211). Statistics Norway.

Shnitser, M. (2025). *The impact of metro accessibility on residential property values and population clustering: A spatial analysis of Washington, DC. The Capitol Economics Journal (peer-reviewed by student editorial board)*, 3(1), 22–42. George Washington University.

Shnitser, M. & Haggerty, E. (2025, April) *The Impacts of Climate Change on Coastal Erosion and Permafrost Degradation Along the Kara Sea*. Poster presented at the *GWU CCAS Research Showcase*, Washington, DC

Finalist, Hsieh Prize for Best Economics Paper in Capstone Course, Department of Economics, George Washington University (2025)

George Washington University Presidential Academic Scholarship, 2021–2025

PROFESSIONAL EXPERIENCE

National Audubon Society	Washington, DC
<i>Dangermond Fellow</i>	<i>June 2026 - Present</i>
- Collaborate with science, policy, and technology stakeholders to design, develop, and manage geospatial products that support conservation planning, species monitoring, and long-term research initiatives - Develop Python- and ArcPy-based geoprocessing tools and scripts to automate recurring spatial analysis, reducing manual processing time and improving reproducibility across projects - Build and maintain ArcGIS Online applications, dashboards, and workflows that support community engagement, data collection, and organization-wide data management - Administer Audubon’s enterprise ArcGIS Online environment for 800+ users, managing user roles, licensing, and platform governance while delivering technical support and training to staff and partners	
TeleGeography	Washington, DC
<i>Market Research and Pricing Analysis Intern</i>	<i>May 2024 - August 2024</i>
- Collected, validated, and updated data for over 7,000 business broadband plans, ensuring data accuracy and consistency within TeleGeography’s database - Conducted quantitative and qualitative market analysis to identify pricing structures, competitive patterns, and regional differences across international telecommunications markets	
United States Senate Veterans Affairs Committee	Washington, DC
<i>Legislative Intern</i>	<i>May 2023 - August 2023</i>
- Analyzed historical data on veteran healthcare access to evaluate the impacts of past policy decisions - Synthesized policy research, hearing materials, and constituent correspondence into concise memos, supporting staff in preparing for briefings and legislative meetings	

LEADERSHIP & SERVICE

Plast Ukrainian Scouting Organization	Philadelphia, PA / Washington, DC
<i>Youth Counselor</i>	<i>September 2018 - Present</i>
Designed, organized, and led weekly educational and service-oriented programs for youth aged 12–16, fostering leadership skills, teamwork, and community engagement	
<i>Philadelphia Branch Lead</i>	<i>May 2025 - May 2026</i>
Oversee branch operations for 100+ members, including program planning, coordinating volunteer recruitment and activities, and community outreach initiatives to expand organizational impact	

- Serve as liaison between branch members and the larger organization, promoting organizational goals and implementing strategies to expand participation and engagement across the organization

GW Humanitarian Mapping Society (HMS)

Washington, DC

Volunteer

January 2022 - May 2025

- Participated in and advanced collaborative mapping projects using OpenStreetMap data, producing high-quality geospatial information to support humanitarian response and disaster relief operations
- Trained and mentored new volunteers in mapping techniques and quality control protocols, enhancing overall project accuracy and increasing volunteer engagement and retention.

GW TRAILS

Washington, DC

Outdoor Guide (Volunteer Position)

March 2022 - May 2025

- Led multi-day backpacking/hiking trips for GW students, managing group logistics and safety in backcountry settings